

MATH70102 UNSUPERVISED LEARNING

LIVE SESSION — WEEK 1

Mikko Pakkanen

Department of Mathematics
Imperial College London

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IMPERIAL

Housekeeping

Review of this week's content

Problem sheet questions

About me

- At [Imperial](#) since 2014, apart from academic year 2022–2023 at the [University of Waterloo](#), Canada
- Before that, postdoc at [Aarhus University](#), Denmark
- Undergrad and PhD at the [University of Helsinki](#), Finland

Teaching at Imperial:

- Previously, in [MSc Maths and Finance](#) and [MSc Stats](#); currently [MLDS](#) and [MSc Stats](#).

Research interests:

- Stochastic processes
- Machine learning and deep learning
- Quantitative finance
- Mathematical modelling of epidemics

Schedule

- Live session:** 9:00–10:00am or 3:00–4:00pm
(British Summer Time)
Wednesdays [recorded & available afterwards]
- Office hour:** 3:00–4:00pm or 9:00–10:00am
(British Summer Time)
Wednesdays [not recorded]

Content outline

1. Dimensionality reduction:

- principal component analysis (PCA)
- extensions of PCA (sparse, kernel, autoencoder)
- non-negative matrix factorisation
- canonical correlation analysis
- independent component analysis
- multidimensional scaling
- non-linear dimensionality reduction methods

2. Density estimation:

- parametric density estimation
- estimation of mixtures using EM algorithm
- non-parametric density estimation (kernel and histogram)

Content outline (cont'd)

3. Clustering:

- k -means clustering
- k -medoids clustering
- hierarchical clustering
- evaluation measures

4. Anomaly and outlier detection:

- local outlier factor
- one-class support vector machine

Literature

The **lecture notes** are designed to be **self-contained**, but for further reading, one can optionally consult the following **two textbooks** which the module is partly based on:

1. C. M. Bishop (2006): *Pattern Recognition and Machine Learning*. Springer, New York.
2. T. Hastie, R. Tibshirani and J. Friedman (2009): *The Elements of Statistical Learning: Data Mining, Inference, and Prediction, 2nd edition*. Springer, New York.

Additionally, references to **original research papers** are provided in lecture notes.

Assessment

The module is assessed through **two pieces of coursework** completed individually, both worth 50%:

- **Coursework 1** — release: 20 May 2026, 9:00am; due: 3 June 2026, 9:00am (British Summer Time),
- **Coursework 2** — release: 1 July 2026, 9:00am; due: 15 July 2026, 9:00am (British Summer Time).

Coursework 1 will be about the topics covered up to that point, whilst Coursework 2 is cumulative, containing tasks on the entire module. Summative **feedback** will be provided afterwards.

Programming languages

Examples will be given using **Python** and **scikit-learn**.

However, you may complete the coursework using whichever language you prefer.

Mathematical prerequisites

1. Linear algebra:

- vectors
- matrices
- linear (in)dependence
- eigenvalues and eigenvectors
- matrix inverses and determinants

2. Multivariate calculus:

- gradients
- optimisation (under linear constraints)

3. Probability:

- random vectors, expectations, covariances
- multivariate distributions

Structure of live sessions

As a rule, the sessions will follow this structure:

1. Review of the current week's **theoretical and methodological content** (+possibly discussion of some material not covered in videos)
2. Computational **examples** (Jupyter notebooks)
3. Solutions to **problem sheet questions**
4. Room for **questions**

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Problem sheet questions

Curse of dimensionality

Data

Suppose we study **samples** $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$

In modern applications of data science, samples are often **high-dimensional** (d is large):

- **time series** ($d \sim$ thousands and more)
- **images** ($d \sim$ hundreds of thousands and more)
- **audio** ($d \sim$ hundreds of thousands and more)
- **video** ($d \sim$ millions and more)

(once flattened to vectors)

Curse of dimensionality (cont'd)

Problems when d is large

1. Visualisation:

- Hard to visualise **dependence** when $d > 2$ (we can study **two-dimensional projections**, but they may not reveal everything)
- Hard to visualise **distribution** when $d > 2$ (**histograms** and **kernel density estimates** can be plotted only up to $d = 2$)

2. Storage (especially if also n is large):

- **Hard drives** and **solid state drives** are nowadays **cheap** (fast 1 TB solid state drive price < £100)
- But **memory** (RAM) can still be a bottleneck
- Problem if data set does not fit in RAM (most efficient data structures require contiguous memory)

Curse of dimensionality (cont'd)

Problems when d is large

3. Computational cost:

- Modern CPUs and GPUs are remarkably **powerful** and **parallelisable** (multiple threads/cores, SIMD)
- But in many situations **computational complexity** grows fast (polynomially)
- For example, **multiplication** of dense $d \times d$ matrices requires $O(d^{2.8})$ operations (using most efficient algorithms available today)

4. Measuring distances:

- Usual metrics (e.g., **Euclidean distance**) may lose relevance in high dimensions

5. Fundamental limits:

- **Mathematics** has its own limits...(example to follow)

Example: Sample covariance matrix

The **sample covariance matrix** of the data set $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$ is

$$\Sigma := \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T,$$

where $\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \in \mathbb{R}^d$ is the sample mean.

Defining

$$X := [\mathbf{x}_1 - \bar{\mathbf{x}} \quad \dots \quad \mathbf{x}_n - \bar{\mathbf{x}}] \in \mathbb{R}^{d \times n},$$

we can write

$$\Sigma = \frac{1}{n} X X^T.$$

Example: Sample covariance matrix (cont'd)

Recall that the **rank** of a matrix $A \in \mathbb{R}^{d \times e}$, denoted $\text{rank}(A)$, tells us **how many linearly independent rows and columns** A has. Key properties:

- $\text{rank}(A) \leq \min\{d, e\}$
- $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$ for conformable A and B
- in the case $d = e$, A is invertible $\Leftrightarrow \text{rank}(A) = d$

Now, if $d > n$, then

$$\text{rank}(\Sigma) = \text{rank}(XX^T) \leq \min\{\text{rank}(X), \text{rank}(X^T)\} \leq \min\{d, n\} = n < d.$$

Hence, Σ^{-1} (known as the **sample precision matrix**) **does not exist**. But some statistical methods rely on the sample precision matrix.

Dimensionality reduction

“Straightforward” ways to reduce dimensionality

- Remove **irrelevant variables**, variables that carry **little or no information** about the phenomenon we are studying
- Remove **redundant variables**, which carry information that is **already conveyed** by other variables in the data set

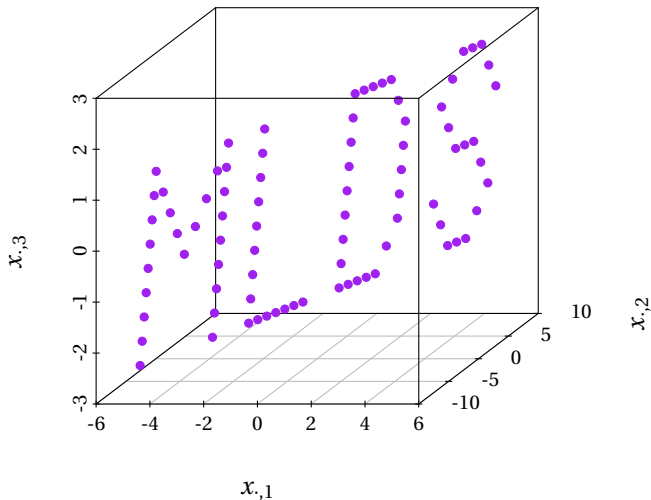
But it is **not** always easy to identify which variables are irrelevant or redundant!

Dimensionality reduction (cont'd)

Intrinsic dimensionality

- In dimensionality reduction, we seek to determine the **minimal set of variables** that describe the essential properties of the data — these variables may be **latent**
- The number of these variables is the **intrinsic dimensionality** of the data set
- Note, however, the word “essential” — this is **not** a rigorous mathematical concept (as it stands)
- We can try to **reconstruct** the original data from this smaller set of variables (how well?)
- Similar to finding the **best compression** of the data

Dimensionality reduction (cont'd)



Dimensionality reduction (cont'd)

Dimensionality reduction methods

- Linear or non-linear
- Supervised or unsupervised
- Suitability for different data types (continuous, discrete, categorical etc) may vary

Our starting point (next week) is **principal component analysis** (PCA), which is a linear, unsupervised dimensionality reduction method

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Problem 1

Question

Can you find concrete examples of applications of data science, machine learning or statistics, where the **curse of dimensionality** appears?

Problem 1

Solution

Example: **variance-optimal investing into stocks** (using daily price data)

- d (number of stocks) **several hundred**
- ~ 250 observations per year, **cannot** go too many years into the past
- Hence, it is not unusual that **$d > n$**
- However, the optimal choice requires the **sample precision matrix** Σ^{-1} , **which does not exist**

Problem 2

Question

Suppose, hypothetically, that we are analysing a data set measuring the **monthly demand** for **each product in a supermarket** over five years.

(Although unrealistic, assume for simplicity that the selection of products remains unchanged throughout the observation period.)

Give examples of potential **latent variables** governing the data.

Problem 2

Solution

- The dimensionality of the data is **enormous**
- However, it is likely that people's consumption choices are **not** so “high-dimensional”
- It is thus plausible that there are latent factors describing demand at the level of **product classes** (fruits, vegetables, dairy products, meat etc)
- Within a product class, the demand for each **individual product** is likely to follow the factor quite closely